

Power Analysis for Longitudinal Clinical Trials with Run-In and Common Close Observations

Ronald G. Thomas, Ph.D.*

July 01, 2026

Abstract

Background. Longitudinal clinical trials often incorporate pre-randomisation (run-in) and post-treatment (common close) observations to improve estimator efficiency. Closed-form power and sample-size formulas for these augmented designs under linear mixed-effects models with random slopes have remained scattered across the literature and partially specified, making applied sample-size calculation cumbersome.

Methods. We derive closed-form power and sample-size formulas for three nested cases: (i) two run-in observations, (ii) an arbitrary number J_0 of run-in observations, and (iii) J_0 run-in plus J_2 common close observations. All derivations proceed via the Woodbury matrix identity applied to the GLS estimator under change-score parameterisation. The formulas are expressed in terms of the residual variance σ^2 , the random-slope variance σ_b^2 , and the observation spacing t , enabling direct computation of required sample sizes.

Results. In a numerical example using parameters from Alzheimer's disease neuroimaging trials, for which the signal-to-noise ratio $\sigma_b/\sigma \approx 1.7$ is favourable, two run-in observations reduce the required sample size by 62% (from 385 to 145 per group), and adding two common close observations raises the reduction to 82%. The magnitude of the gain is governed by σ_b/σ : across a representative range of slope variability the reduction from two run-in observations spans roughly 14% at low ratios to 76% at high ratios. The closed-form expressions allow this sensitivity analysis with respect to σ_b/σ without re-running simulation.

Conclusions. The Woodbury-based formulas are the simplest available closed-form expressions for the variance of the treatment-effect estimator under random-slopes longitudinal-trial designs with run-in and common close observations. They serve as a foundation for the AR(1) extension (Paper 2), the optimal-allocation problem (Paper 3), and the GLS-versus-ANCOVA-versus-averaged estimator comparison (Paper 4) in this compendium.

Contents

1 Introduction

2

*Wertheim School of Public Health, UCSD; ORCID: 0000-0003-1686-4965

36	2 Model	5
37	2.1 General formulation	5
38	2.2 Change-score parameterization	8
39	2.3 Covariance structure	8
40	3 Variance of the Treatment Effect Estimator	9
41	3.1 GLS estimation and the Woodbury identity	9
42	3.2 Case 1: Standard design, no run-in ($J_0 = 0, J_1 = 2, J_2 = 0$)	11
43	3.3 Case 2: Two run-in observations ($J_0 = 2, J_1 = 1, J_2 = 0$)	12
44	3.4 General formula: J_0 run-in observations ($J_2 = 0$)	16
45	3.5 General formula with common close ($J_2 \geq 0$)	20
46	4 Power and Sample Size Formulas	22
47	5 Numerical Examples	23
48	5.1 Verification of closed-form results	23
49	5.2 Sample size as a function of run-in observations	26
50	5.3 Sample size as a function of common close observations	26
51	5.4 Relative efficiency surface	26
52	5.5 Application: Alzheimer's disease imaging trial	26
53	6 Discussion	26
54	7 Conflict of interest	29
55	8 Funding	29
56	9 Data availability statement	29
57	10 Reproducibility	29
58	11 References	30

59 1 Introduction

60 [1] **Beyond its traditional roles, the run-in period has a dis-**
61 **distinct methodological value in rate-of-change trials: estimat-**
62 **ing individual baseline trajectories to raise power.**

- 63 • Conventional run-in uses include identifying compliant partici-
64 pants and washing out prior treatments.
- 65 • When the outcome is a rate of change, run-in observations esti-
66 mate each participant's baseline trajectory.
- 67 • Incorporating those trajectories into the analysis model substan-
68 tially increases statistical power.

69 To be completed by rgt.

The run-in period has served multiple purposes in clinical trial design, from identifying compliant participants to washing out prior treatments¹. A distinct and methodologically important use has emerged in trials where the primary outcome is a rate of change: estimating individual baseline trajectories that, when incorporated into the analysis model, substantially increase statistical power.

[2] Frost and colleagues established the mixed-model foundation: run-in observations help most when slope variability is large relative to measurement error.

- The key insight is that pre-randomization observations inform each participant’s underlying trajectory.
- This reduces the residual variance of the treatment-effect estimator, and the gain grows with the slope-to-noise ratio.
- In Alzheimer’s neuroimaging trials, where that ratio is favorable, they reported sample-size reductions of 30 to 45 percent.

To be completed by rgt.

² established the theoretical foundation for this approach within the linear mixed effects framework. Their key insight was that when between-subject variability in rates of change is large relative to measurement error, pre-randomization observations provide valuable information about each participant’s underlying trajectory, reducing the residual variance of the treatment effect estimator. Working with neuroimaging outcomes in Alzheimer’s disease trials, where the ratio of slope variability to measurement error is particularly favorable, they demonstrated sample size reductions of 30–45%.

[3] Wang and colleagues extended the framework with two-period models that jointly model run-in and post-randomization data rather than treating run-in as a covariate.

- Joint modelling of both periods yields additional power gains of up to 15 percent.
- It permits unequal randomization ratios without power loss.
- Run-in observations contribute placebo-equivalent information for all participants.

To be completed by rgt.

³ extended this framework by proposing two-period linear mixed effects models that jointly model run-in and post-randomization data rather than using run-in observations merely as covariates. Their approach yields additional power gains of up to 15% and enables unequal randomization ratios without power loss, since run-in observations effectively contribute placebo-equivalent information for all participants.

[4] A parallel literature gives closed-form variance and sample-size formulae for the random-coefficient model without a run-in phase, which form the baseline this paper

extends.

- Lu and colleagues derived the cLDA estimator variance, its link to ANCOVA, and the conditions under which the two coincide.
- Hu and colleagues generalized these to random-coefficient models with monotone missing data and variable follow-up, illustrated with ADNI data.
- These results cover only the $J_0 = 0$ case and serve as the baseline extended here.

To be completed by rgt.

A parallel literature has developed closed-form variance and sample size formulae for the random coefficient regression model without a run-in phase.⁴ derived the variance of the constrained longitudinal data analysis estimator and its relation to analysis of covariance, and⁵ established efficiency conditions under which the two approaches coincide.⁶ generalized these results to random coefficient regression models with monotone missing data and variable follow-up, providing variance, power, and sample size expressions illustrated with data from the Alzheimer’s Disease Neuroimaging Initiative. These formulae apply to designs with $J_0 = 0$ run-in observations and an arbitrary post-randomization visit schedule, and serve as the baseline case that we extend here.

[5] **The CRAN longpower package is the reference implementation of the baseline formulae, and our general result reduces to its routines at the appropriate boundaries.**

- longpower provides edland, diggle, and liu.liang routines for slope-comparison trials.
- Our formula reduces to the Ard-Edland slope variance at $J_0 = J_2 = 0$, and to the Diggle compound-symmetry form when also $\sigma_b^2 = 0$.
- The generalization thus embeds the existing longpower formulae as the baseline against which run-in and common-close gains are measured.

To be completed by rgt.

The CRAN reference implementation of the $J_0 = 0$ formulae is the longpower package⁷, which provides the routines `edland.linear.power()`, `diggle.linear.power()`, and `liu.liang.linear.power()` for slope-comparison trials with random intercepts and slopes^{8–10}. We show in Section 3.3 that the closed-form result derived below reduces to the Ard-Edland slope-variance formula implemented in `edland.linear.power()` at the boundary $J_0 = J_2 = 0$, and to the Diggle compound-symmetry formula when additionally $\sigma_b^2 = 0$. The generalization developed here therefore embeds the existing longpower formulae as the baseline case against which the efficiency gains from run-in and common close observations are measured.

[6] **No prior closed-form power formula handles an arbitrary number of run-in observations together with a separate**

common-close phase, the gap this paper fills.

- The common-close phase observes all participants off-treatment after the randomized period.
- It supplies extra information about individual trajectories at no cost to the treatment comparison.
- Such post-treatment periods appear in delayed-start and randomized-withdrawal designs, but their GLS efficiency contribution has not been formally characterized.

To be completed by rgt.

Despite these advances, no general closed-form power formula has been derived that simultaneously accounts for an arbitrary number of run-in observations and a separate set of common close observations collected after treatment cessation. The common close phase, in which all participants are observed off-treatment following the randomized period, provides additional information about individual trajectories at no cost to the treatment comparison. Post-treatment observation periods appear in delayed-start and randomized withdrawal designs aimed at distinguishing symptomatic from disease-modifying effects¹¹, but the contribution of the common close to statistical efficiency within the GLS framework has not been formally characterized.

[7] **The paper presents three nested results, all derived from the GLS estimator via the Woodbury identity.**

- A closed-form variance for exactly two run-in observations (Section 3.2), then its generalization to arbitrary J_0 (Section 3.3).
- A further extension adding J_2 common-close observations (Section 3.4).
- All expressions are in terms of σ^2 , σ_b^2 , and spacing t , with numerical examples from Alzheimer’s neuroimaging studies (Section 5).

To be completed by rgt.

We present three results. First, we derive a simple closed-form variance formula for the treatment effect estimator when exactly two run-in observations precede randomization (Section 3.2). Second, we generalize this formula to an arbitrary number J_0 of run-in observations (Section 3.3). Third, we further extend to include J_2 common close observations collected after treatment cessation (Section 3.4). All derivations proceed from the generalized least squares estimator under the Woodbury matrix identity, yielding expressions in terms of the residual variance σ^2 , the random slope variance σ_b^2 , and the observation spacing t . Section 5 provides numerical examples using parameters estimated from Alzheimer’s disease neuroimaging studies.

2 Model

2.1 General formulation

[8] **The model accommodates a two-arm longitudinal trial**

whose observation schedule spans three distinct phases.

- N participants are observed at equally spaced time points.
- The phases are run-in (pre-randomization), treatment (post-randomization), and an optional common close (post-treatment).
- Y_{ij} denotes the outcome for participant i at time point j .

To be completed by rgt.

Consider a two-arm longitudinal clinical trial in which N participants are observed at equally spaced time points across three phases: a run-in period (pre-randomization), a treatment period (post-randomization), and an optional common close period (post-treatment). Let Y_{ij} denote the outcome for participant i at time point j . The observation schedule is:

- **Run-in:** $j = -J_0, -J_0 + 1, \dots, -1$ (all participants untreated)
- **Randomization visit:** $j = 0$ (baseline)
- **Treatment period:** $j = 1, 2, \dots, J_1$ (participants randomized to treatment or placebo)
- **Common close:** $j = J_1 + 1, J_1 + 2, \dots, J_1 + J_2$ (all participants off-treatment)

We model the outcome as

$$Y_{ij} = \alpha + a_i + (\delta(1 - h_j) + \beta h_j + \gamma h_j g_i + b_i)t_j + u_{ij} + \varepsilon_{ij} \quad (1)$$

[9] **The outcome model combines a random intercept and random slope with a three-component within-subject variance structure following Frost and colleagues.**

- a_i and b_i are the random intercept and slope, while u_{ij} and ε_{ij} are the between-visit and between-scan residuals.
- Both residual terms are independent across visits, so under a single scan per visit they enter every formula only through their sum.
- We therefore define the total within-visit residual $w_{ij} = u_{ij} + \varepsilon_{ij}$ with variance σ^2 .

To be completed by rgt.

where α is the overall intercept, $a_i \sim N(0, \sigma_a^2)$ is a random intercept, $b_i \sim N(0, \sigma_b^2)$ is a random slope, $u_{ij} \sim N(0, \sigma_u^2)$ are within-subject between-visit random effects independent across visits, and $\varepsilon_{ij} \sim N(0, \sigma_\varepsilon^2)$ are independent between-scan measurement errors. This three-component within-subject variance structure follows² (their Eq. 17). Because u_{ij} and ε_{ij} are both independent across visits, under a single scan per visit they enter every formula below only through their sum. We therefore write $w_{ij} = u_{ij} + \varepsilon_{ij}$ for the total within-visit residual, with

$$\sigma^2 \equiv \text{Var}(w_{ij}) = \sigma_u^2 + \sigma_\varepsilon^2. \quad (2)$$

[10] **The two within-visit components are identifiable only**

235 with replicate scans, so single-scan designs depend solely on
236 their sum.

- 237 • Separating σ_u^2 from σ_ε^2 requires a visit carrying replicate scans.
- 238 • For the single-scan designs treated here, only the sum σ^2 enters
239 the treatment-effect variance.
- 240 • This reduction simplifies all subsequent variance expressions.

241 To be completed by rgt.

242 *The two within-visit components are separately identifiable only when a visit*
243 *carries replicate scans; for the single-scan designs treated here only their sum*
244 *σ^2 enters the variance of the treatment-effect estimator. The phase indicator*
245 *is*

$$h_j = \begin{cases} 0 & \text{if } j \leq 0 \quad (\text{run-in}) \\ 1 & \text{if } j > 0 \quad (\text{post-randomization}) \end{cases}$$

246 and the treatment group indicator is

$$g_i = \begin{cases} 1 & i = 1, \dots, m \quad (\text{treatment}) \\ 0 & i = m + 1, \dots, N \quad (\text{placebo}) \end{cases}$$

247 [11] Three fixed-effect slope parameters describe the run-in,
248 placebo post-randomization, and treatment trajectories un-
249 der equally spaced observations.

- 250 • δ is the run-in rate of change common to all participants.
- 251 • β is the placebo post-randomization rate of change, and γ is the
252 treatment effect on that rate.
- 253 • Observations are equally spaced with interval t , so $t_j = jt$.

254 To be completed by rgt.

255 *The fixed effects are: δ , the rate of change during the run-in period (common*
256 *to all participants); β , the rate of change during the post-randomization period*
257 *for the placebo group; and γ , the treatment effect on the rate of change. We*
258 *assume equally spaced observations with interval t , so $t_j = jt$.*

259 [12] During the common close both arms revert to a shared
260 off-treatment slope, modelled by deactivating the treatment
261 indicator in that phase.

- 262 • For $j > J_1$ all participants are off-treatment with slope $\beta + b_i$.
- 263 • This is implemented by setting $g_i = 0$ for every participant during
264 the common close.
- 265 • Equivalently, the treatment indicator is replaced with $g_i \cdot I(1 \leq$
266 $j \leq J_1)$ in the model.

267 To be completed by rgt.

For the common close phase ($j > J_1$), both treatment and placebo groups revert to a common off-treatment slope. We model this by setting $g_i = 0$ for all participants during the common close, so that the slope for both groups is $\beta + b_i$ in this phase. This is equivalent to replacing the treatment indicator with $g_i \cdot I(1 \leq j \leq J_1)$ in model (1).

2.2 Change-score parameterization

Following², we work with change scores relative to the randomization visit:

$$c_{ij} = Y_{ij} - Y_{i0} \quad (3)$$

[13] **Differencing relative to baseline removes the intercept and aligns the analysis with constrained longitudinal data analysis under compound symmetry.**

- The change score eliminates α and the random intercept a_i , leaving only two variance components.
- It is algebraically equivalent to cLDA, so the derived formulas apply to both analyses of run-in designs.
- By dropping σ_a^2 it avoids the anticonservative sample sizes that arise from ignoring intercept-slope correlation.

To be completed by rgt.

This eliminates the intercept α and the random intercept a_i . We note that the change-score approach is algebraically equivalent to constrained longitudinal data analysis (cLDA) under compound symmetry, where the baseline is included in the response vector with a common-mean constraint across treatment groups^{12,13}. The variance formulas derived below therefore apply to both change-score and cLDA analyses of run-in designs. Because the change-score eliminates a_i , the resulting formulas depend on only two variance components (σ^2, σ_b^2) rather than the three ($\sigma^2, \sigma_a^2, \sigma_b^2$) required by the raw-outcome model.¹⁴ show that ignoring the intercept-slope correlation in the three-component model can produce anticonservative sample sizes; the change-score approach avoids this issue by construction, at the cost of discarding information about σ_a^2 .

The change-score formulation yields

$$c_{ij} = (\delta(1 - h_j) + \beta h_j + \gamma h_j g_i + b_i) t_j + (w_{ij} - w_{i0}) \quad (4)$$

for $j \neq 0$. The change-score vector for participant i has dimension $J = J_0 + J_1 + J_2$, encompassing all three phases.

2.3 Covariance structure

The marginal covariance of the change-score vector is

$$\Sigma = R + ZGZ' \quad (5)$$

where $G = \sigma_b^2$ (scalar, since we have a single random slope), $Z = t \cdot \mathbf{1}_J$ is the $J \times 1$ random effects design vector, and R is the $J \times J$ residual covariance of the differenced within-visit residuals ($w_{ij} - w_{i0}$).

The residual covariance has the structure

$$R_{jl} = \text{Cov}(w_{ij} - w_{i0}, w_{il} - w_{i0}) = \begin{cases} 2\sigma^2 & \text{if } j = l \\ \sigma^2 & \text{if } j \neq l \end{cases} \quad (6)$$

[14] **The shared baseline residual induces compound-symmetry correlation among change scores, so the residual covariance depends only on the summed within-visit variance.**

- The within-visit residuals are independent across visits with common variance σ^2 .
- Differencing against the shared baseline w_{i0} produces positive correlation, giving $R = \sigma^2(\mathbf{I}_J + \mathbf{1}_J\mathbf{1}_J')$.
- Both within-visit components enter R identically, so only their sum is identified from single-scan designs.

To be completed by rgt.

since $w_{ij} = u_{ij} + \varepsilon_{ij}$ are independent across visits with common variance $\sigma^2 = \sigma_u^2 + \sigma_\varepsilon^2$, and the shared baseline residual w_{i0} induces positive correlation among all change scores. Thus $R = \sigma^2(\mathbf{I}_J + \mathbf{1}_J\mathbf{1}_J')$. Both within-visit components enter R identically because u_{i0} and ε_{i0} are differenced out of every change score in the same way; this is why only their sum σ^2 is identified from single-scan designs.

3 Variance of the Treatment Effect Estimator

3.1 GLS estimation and the Woodbury identity

The generalized least squares estimator of the fixed effects is

$$\hat{\beta} = (X'\Sigma^{-1}X)^{-1}X'\Sigma^{-1}\mathbf{c} \quad (7)$$

[15] **The estimand of interest is the treatment-effect variance, a single diagonal element of the inverted GLS information matrix.**

- The GLS estimator has covariance $(X'\Sigma^{-1}X)^{-1}$.
- X is the fixed-effects design matrix and $\mathbf{c}$ is the stacked change-score vector.
- The target $\text{Var}(\hat{\gamma})$ is the diagonal entry corresponding to γ .

To be completed by rgt.

with $\text{Var}(\hat{\beta}) = (X'\Sigma^{-1}X)^{-1}$, where X is the fixed effects design matrix and $\mathbf{c}$ is the stacked change-score vector. Our target is $\text{Var}(\hat{\gamma})$, the variance of

336 the treatment effect estimator, which is the diagonal element of $(X'\Sigma^{-1}X)^{-1}$
 337 corresponding to γ .

338 The key computational tool is the Woodbury matrix identity. Since $\Sigma = R + ZGZ'$
 339 with scalar $G = \sigma_b^2$:

$$\Sigma^{-1} = R^{-1} - R^{-1}Z(G^{-1} + Z'R^{-1}Z)^{-1}Z'R^{-1} \quad (8)$$

340 This avoids direct inversion of the $J \times J$ matrix Σ and enables closed-form results
 341 because R^{-1} has a known simple structure.

342 From $R = \sigma^2(\mathbf{I}_J + \mathbf{1}_J\mathbf{1}_J')$, the Sherman-Morrison formula gives

$$R^{-1} = \frac{1}{\sigma^2} \left(\mathbf{I}_J - \frac{1}{J+1} \mathbf{1}_J \mathbf{1}_J' \right) \quad (9)$$

343 With $Z = t \cdot \mathbf{1}_J$, we obtain

$$Z'R^{-1}Z = \frac{t^2}{\sigma^2} \left(J - \frac{J^2}{J+1} \right) = \frac{Jt^2}{\sigma^2(J+1)} \quad (10)$$

344 and

$$(G^{-1} + Z'R^{-1}Z)^{-1} = \frac{\sigma^2\sigma_b^2(J+1)}{\sigma^2(J+1) + Jt^2\sigma_b^2} \quad (11)$$

345 Let $a = \sigma^2$ and $b = \sigma_b^2$ for conciseness. Define

$$\eta_J = \frac{a(J+1) + Jt^2b}{a(J+1)} = 1 + \frac{Jt^2b}{a(J+1)} \quad (12)$$

346 so that $(G^{-1} + Z'R^{-1}Z)^{-1} = b/\eta_J$.

347 **3.1.1 Per-participant information, compact form**

348 Let X_i , Z_i , and R_i denote the per-participant fixed effects design, random effects
 349 design, and residual covariance, and define the scalar

$$\mathcal{J}_i = Z_i'R_i^{-1}Z_i + G^{-1} \quad (13)$$

350 By (8), the per-participant contribution to the GLS information matrix is

$$X_i'\Sigma_i^{-1}X_i = X_i'R_i^{-1}X_i - \frac{(X_i'R_i^{-1}Z_i)(Z_i'R_i^{-1}X_i)}{\mathcal{J}_i} \quad (14)$$

351 [16] **The additive per-participant information form has two**
 352 **properties exploited later: a scalar rank-one correction and**
 353 **accommodation of heterogeneous designs.**

- 354 • The total information is the sum of per-participant contributions,
 355 with covariance given by its inverse.

- Under a single random slope, $\mathcal{J}_i$ is a scalar, making the correction a rank-one adjustment of the OLS-style information.
- Because contributions add, varying J_0 , J_2 , or dropout patterns are handled by summing over strata.

To be completed by rgt.

and the total information is $X'\Sigma^{-1}X = \sum_i X_i'\Sigma_i^{-1}X_i$, with $\text{Var}(\hat{\beta}) = (X'\Sigma^{-1}X)^{-1}$. Two features of this form are used later: (i) under a single random slope, $\mathcal{J}_i$ is a scalar so (14) is a rank-one correction of the OLS-style information $X_i'R_i^{-1}X_i$; (ii) because per-participant contributions add, heterogeneous designs (varying J_0 , J_2 , or dropout patterns) are accommodated by summing (14) over the relevant strata.

3.2 Case 1: Standard design, no run-in ($J_0 = 0$, $J_1 = 2$, $J_2 = 0$)

[17] **The no-run-in case reproduces established results and serves as a verification baseline for the framework.**

- It reproduces Frost and colleagues, Section 3.1.1.
- It is the $J_0 = 0$ baseline of the random-coefficient power formulae of Lu and Hu and colleagues.
- It thereby provides a verification checkpoint for the present derivation.

To be completed by rgt.

This case reproduces the result in², Section 3.1.1, and is the $J_0 = 0$ baseline of the random coefficient regression power formulae of⁴ and⁶. It serves as verification for our framework.

[18] **Without a run-in phase the design reduces to two post-randomization change scores and a two-parameter fixed-effects model.**

- There are $J = 2$ change scores at post-randomization times t and $2t$.
- The absence of run-in removes the δ parameter from the model.
- The remaining fixed effects are the placebo slope β and the treatment effect γ .

To be completed by rgt.

With $J_0 = 0$ and $J_1 = 2$, there are $J = 2$ change scores $(c_{i,1}, c_{i,2})$ at post-randomization times $(t, 2t)$. Since there is no run-in, the model has two fixed effects: the placebo slope β and the treatment effect γ .

The design matrices are:

$$X_i^{(\text{trt})} = t \begin{pmatrix} 1 & 1 \\ 2 & 2 \end{pmatrix}, \quad X_i^{(\text{plc})} = t \begin{pmatrix} 1 & 0 \\ 2 & 0 \end{pmatrix}$$

392 The residual covariance is $R = a \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$ with $R^{-1} = \frac{1}{3a} \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}$, and the
 393 random effects design is $Z = t(1, 2)'$.

394 Applying the Woodbury identity with $J = 2$:

$$Z'R^{-1}Z = \frac{t^2}{3a}(1, 2) \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix} \begin{pmatrix} 1 \\ 2 \end{pmatrix} = \frac{t^2}{3a}(1, 2)(0, 3)' = \frac{6t^2}{3a} = \frac{2t^2}{a} \quad (15)$$

395

$$(G^{-1} + Z'R^{-1}Z)^{-1} = \frac{ab}{a + 2bt^2} \quad (16)$$

396 After summing over one treatment and one placebo participant, the information
 397 matrix is

$$X'\Sigma^{-1}X = \frac{2t^2}{a + 2bt^2} \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix} \quad (17)$$

398 and its inverse is

$$\text{Var}(\hat{\beta}) = \frac{a + 2bt^2}{2t^2} \begin{pmatrix} 1 & -1 \\ -1 & 2 \end{pmatrix} \quad (18)$$

399 The variance of the treatment effect estimator (per participant pair) is the (2, 2)
 400 element:

$$\boxed{\text{Var}(\hat{\gamma})|_{J_0=0, J_1=2} = \frac{a + 2bt^2}{t^2} = \frac{\sigma^2 + 2\sigma_b^2 t^2}{t^2}} \quad (19)$$

401 **[19] The recovered variance matches Frost and supplies the**
 402 **per-group baseline for all run-in comparisons.**

- 403 • The expression agrees with Frost and colleagues, Section 3.1.1.
- 404 • For m participants per group the variance scales as $(\sigma^2 +$
 405 $2\sigma_b^2 t^2)/(mt^2)$.
- 406 • This quantity is the reference against which run-in designs are
 407 evaluated.

408 *To be completed by rgt.*

409 *This agrees with Frost et al. (2008), Section 3.1.1. For m participants per*
 410 *group: $\text{Var}(\hat{\gamma}) = (\sigma^2 + 2\sigma_b^2 t^2)/(mt^2)$. This serves as the baseline against*
 411 *which the run-in designs are compared.*

412 3.3 Case 2: Two run-in observations ($J_0 = 2, J_1 = 1, J_2 = 0$)

413 **[20] Two run-in observations yield three change scores and a**
 414 **design matrix that distinguishes the run-in slope from the**
 415 **post-randomization slopes.**

- 416 • There are $J = 3$ change scores spanning the pre- and post-
 417 randomization times.
- 418 • The fixed-effects model now includes the run-in slope δ alongside
 419 β and γ .

- The design matrix separates the run-in contribution from the post-randomization contribution.

To be completed by rgt.

We now have $J = 3$ change scores $(c_{i,-2}, c_{i,-1}, c_{i,1})$ at times $(-2t, -t, 0, t)$. The fixed effects design matrix separates the run-in slope δ from the post-randomization slopes:

$$X_i^{(\text{trt})} = \begin{pmatrix} -2t & 0 & 0 \\ -t & 0 & 0 \\ 0 & t & t \end{pmatrix}, \quad X_i^{(\text{plc})} = \begin{pmatrix} -2t & 0 & 0 \\ -t & 0 & 0 \\ 0 & t & 0 \end{pmatrix}$$

[21] Because the random slope acts uniformly across visits, the random-effects design vector collapses to the scaled time distances from baseline.

- The random slope b_i applies to all time points alike.
- The random-effects design therefore reduces to $Z = t(-2, -1, 1)'$.
- The run-in change scores are differences of the pre-randomization outcomes from baseline.

To be completed by rgt.

The random effects design is $Z = t(1, 1, 1)' \cdot (-2, -1, 1)' \cdot b_i$. However, since the random slope applies uniformly to all time points, $Z = t(-2, -1, 1)'$ for the random effect b_i . We note that the change scores at times $-2t$ and $-t$ relative to baseline at 0 yield $c_{i,-2} = Y_{i,-2} - Y_{i,0}$ and $c_{i,-1} = Y_{i,-1} - Y_{i,0}$.

More carefully, under model (1):

$$\begin{aligned} c_{i,-2} &= -2(\delta + b_i)t + (w_{i,-2} - w_{i,0}) \\ c_{i,-1} &= -(\delta + b_i)t + (w_{i,-1} - w_{i,0}) \\ c_{i,1} &= (\beta + \gamma g_i + b_i)t + (w_{i,1} - w_{i,0}) \end{aligned} \tag{20}$$

Thus the design matrices for the fixed effects $(\delta, \beta, \gamma)'$ are:

$$X_i^{(\text{trt})} = t \begin{pmatrix} -2 & 0 & 0 \\ -1 & 0 & 0 \\ 0 & 1 & 1 \end{pmatrix}, \quad X_i^{(\text{plc})} = t \begin{pmatrix} -2 & 0 & 0 \\ -1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}$$

and the random effects design is $Z_i = t(-2, -1, 1)'$.

The residual covariance is

$$R = a \begin{pmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{pmatrix} = a(\mathbf{I}_3 + \mathbf{1}_3 \mathbf{1}_3') \tag{21}$$

442 with inverse

$$R^{-1} = \frac{1}{a} \left(\mathbf{I}_3 - \frac{1}{4} \mathbf{1}_3 \mathbf{1}_3' \right) = \frac{1}{4a} \begin{pmatrix} 3 & -1 & -1 \\ -1 & 3 & -1 \\ -1 & -1 & 3 \end{pmatrix} \quad (22)$$

443 Now we compute the Woodbury components. With $Z = t(-2, -1, 1)'$:

$$Z' R^{-1} Z = \frac{t^2}{4a} (3 \cdot 4 + 3 \cdot 1 + 3 \cdot 1 - 2(-1)(-2)(-1) - 2(-1)(-2)(1) - 2(-1)(-1)(1)) \quad (23)$$

444 Computing directly: $Z' R^{-1} Z = \frac{t^2}{4a} (-2, -1, 1) \begin{pmatrix} 3 & -1 & -1 \\ -1 & 3 & -1 \\ -1 & -1 & 3 \end{pmatrix} (-2, -1, 1)'$

$$R^{-1} Z = \frac{t}{4a} \begin{pmatrix} 3(-2) + (-1)(-1) + (-1)(1) \\ (-1)(-2) + 3(-1) + (-1)(1) \\ (-1)(-2) + (-1)(-1) + 3(1) \end{pmatrix} = \frac{t}{4a} \begin{pmatrix} -6 \\ -2 \\ 6 \end{pmatrix}$$

445 so $Z' R^{-1} Z = \frac{t^2}{4a} [(-2)(-6) + (-1)(-2) + (1)(6)] = \frac{t^2}{4a} (12 + 2 + 6) = \frac{20t^2}{4a} = \frac{5t^2}{a}$.

446 Alternatively, we note $Z = t \cdot \mathbf{d}$ where $\mathbf{d} = (-2, -1, 1)'$, so $Z' R^{-1} Z = \frac{t^2}{a} (\mathbf{d}' \mathbf{d} -$
447 $\frac{1}{4} (\mathbf{1}' \mathbf{d})^2) = \frac{t^2}{a} (6 - \frac{4}{4}) = \frac{5t^2}{a}$.

448 Thus

$$H \equiv (G^{-1} + Z' R^{-1} Z)^{-1} = \frac{ab}{a + 5bt^2} \quad (24)$$

449 For the information matrix, we compute $X' R^{-1} X$ summed over treatment and
450 placebo groups. For a single treatment participant:

$$X_i^{(\text{trt})'} R^{-1} X_i^{(\text{trt})} = \frac{t^2}{4a} \begin{pmatrix} -2 & -1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 3 & -1 & -1 \\ -1 & 3 & -1 \\ -1 & -1 & 3 \end{pmatrix} \begin{pmatrix} -2 & 0 & 0 \\ -1 & 0 & 0 \\ 0 & 1 & 1 \end{pmatrix}$$

451 Computing the product:

$$X_i^{(\text{trt})'} R^{-1} X_i^{(\text{trt})} = \frac{t^2}{4a} \begin{pmatrix} 11 & 3 & 3 \\ 3 & 3 & 3 \\ 3 & 3 & 3 \end{pmatrix} \quad (25)$$

452 For placebo (whose γ column is zero):

$$X_i^{(\text{plc})'} R^{-1} X_i^{(\text{plc})} = \frac{t^2}{4a} \begin{pmatrix} 11 & 3 & 0 \\ 3 & 3 & 0 \\ 0 & 0 & 0 \end{pmatrix} \quad (26)$$

453 Summing (one participant per group for the derivation, then multiplying by m):

$$X'R^{-1}X = \frac{t^2}{4a} \begin{pmatrix} 22 & 6 & 3 \\ 6 & 6 & 3 \\ 3 & 3 & 3 \end{pmatrix} \quad (27)$$

454 For the Woodbury correction, we need $X'R^{-1}Z$. For treatment:

$$X_i^{(\text{trt})'}R^{-1}Z = \frac{t^2}{4a} \begin{pmatrix} -2 & -1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} -6 \\ -2 \\ 6 \end{pmatrix} = \frac{t^2}{4a} \begin{pmatrix} 14 \\ 6 \\ 6 \end{pmatrix}$$

455 For placebo:

$$X_i^{(\text{plc})'}R^{-1}Z = \frac{t^2}{4a} \begin{pmatrix} -2 & -1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} -6 \\ -2 \\ 6 \end{pmatrix} = \frac{t^2}{4a} \begin{pmatrix} 14 \\ 6 \\ 0 \end{pmatrix}$$

456 The Woodbury correction requires the per-subject outer products (not the outer
457 product of the sum):

$$\begin{aligned} W &= \sum_g X_i^{(g)'}R^{-1}Z \cdot H \cdot Z'R^{-1}X_i^{(g)} \\ &= \frac{bt^4}{16a(a+5bt^2)} \left(\begin{pmatrix} 14 \\ 6 \\ 6 \end{pmatrix} (14, 6, 6) + \begin{pmatrix} 14 \\ 6 \\ 0 \end{pmatrix} (14, 6, 0) \right) \\ &= \frac{bt^4}{16a(a+5bt^2)} \begin{pmatrix} 392 & 168 & 84 \\ 168 & 72 & 36 \\ 84 & 36 & 36 \end{pmatrix} \end{aligned} \quad (28)$$

458 Therefore

$$X'\Sigma^{-1}X = X'R^{-1}X - W \quad (29)$$

459 **[22] The treatment-effect variance is the relevant diagonal el-**
460 **ement of the inverted information matrix, obtained numeri-**
461 **cally and checked against the closed form.**

- 462 • $\text{Var}(\hat{\gamma})$ is the $(3, 3)$ element of $(X'\Sigma^{-1}X)^{-1}$.
- 463 • It is evaluated numerically in the accompanying R code.
- 464 • The numerical value is verified against the derived closed-form
- 465 result.

466 To be completed by rgt.

467 *To find $\text{Var}(\hat{\gamma})$, we need the $(3, 3)$ element of $(X'\Sigma^{-1}X)^{-1}$. We compute this*
468 *numerically in the R code below and verify against the closed-form result.*

469 **[23] Completing the cofactor and determinant algebra yields**

the closed-form treatment-effect variance for the two-run-in design.

- Define the auxiliary quantity $D = \sigma^2 + 5\sigma_b^2 t^2$.
- The $(3, 3)$ cofactor simplifies to $6t^4/(aD)$.
- Dividing the cofactor by the determinant gives the boxed variance below.

To be completed by rgt.

Carrying through the algebra, define $D = a + 5bt^2 = \sigma^2 + 5\sigma_b^2 t^2$ and write $X'\Sigma^{-1}X = \frac{t^2}{16aD} A$. The matrix A has $(3, 3)$ minor $1536 aD$ and determinant $9216 aD (a + 2bt^2)$. Hence the $(3, 3)$ cofactor of $X'\Sigma^{-1}X$ is $1536 aD \cdot t^4 / (16aD)^2 = 6t^4 / (aD)$, and its determinant is $9216 aD (a + 2bt^2) \cdot t^6 / (16aD)^3$. The ratio gives:

$$\boxed{\text{Var}(\hat{\gamma})|_{J_0=2, J_1=1} = \frac{8\sigma^2(\sigma^2 + 5\sigma_b^2 t^2)}{3t^2(\sigma^2 + 2\sigma_b^2 t^2)}} \quad (30)$$

[24] The run-in variance differs structurally from the Frost baseline by involving the residual variance in both numerator and denominator.

- In the Frost formula σ^2 appears only additively in the numerator.
- Here σ^2 enters both the numerator and the denominator.
- This reflects the extra information extracted from pre-randomization observations.

To be completed by rgt.

Unlike the Frost formula (19), the run-in variance depends on σ^2 both in the numerator and denominator, reflecting the additional information extracted from pre-randomization observations.

3.4 General formula: J_0 run-in observations ($J_2 = 0$)

[25] The derivation now generalizes to an arbitrary number of run-in observations with no common close.

- The design has arbitrary J_0 run-in and J_1 post-randomization observations.
- The common close is absent, so $J_2 = 0$.
- The total number of change scores is $J = J_0 + J_1$.

To be completed by rgt.

We now derive the variance for arbitrary J_0 run-in observations with J_1 post-randomization observations and no common close ($J_2 = 0$). The total number of change scores is $J = J_0 + J_1$.

504 The change scores are

$$\begin{aligned} c_{i,-s} &= -s(\delta + b_i)t + (w_{i,-s} - w_{i0}) & s &= 1, \dots, J_0 \\ c_{i,l} &= l(\beta + \gamma g_i + b_i)t + (w_{i,l} - w_{i0}) & l &= 1, \dots, J_1 \end{aligned} \quad (31)$$

505 [26] The design matrix is built from run-in and post-
506 randomization time vectors mapping to the three fixed-effect
507 columns.

- 508 • The columns correspond to the parameters δ , β , and γ .
- 509 • $\mathbf{d}_{J_0}$ collects the negative run-in times scaled by t .
- 510 • $\mathbf{d}_{J_1}$ collects the positive post-randomization times scaled by t .

511 To be completed by rgt.

512 The fixed effects design matrix has columns for (δ, β, γ) . Let $\mathbf{d}_{J_0} =$
513 $(-J_0, -J_0 + 1, \dots, -1)' \cdot t$ be the run-in time vector and $\mathbf{d}_{J_1} = (1, 2, \dots, J_1)' \cdot t$
514 be the post-randomization time vector. Then:

$$X_i^{(\text{trt})} = \begin{pmatrix} \mathbf{d}_{J_0} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{d}_{J_1} & \mathbf{d}_{J_1} \end{pmatrix}, \quad X_i^{(\text{ple})} = \begin{pmatrix} \mathbf{d}_{J_0} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{d}_{J_1} & \mathbf{0} \end{pmatrix}$$

515 The random effects design vector is $Z_i = (\mathbf{d}_{J_0}', \mathbf{d}_{J_1}')'$, i.e., the full vector of time
516 distances from baseline.

517 The residual covariance remains $R = a(\mathbf{I}_J + \mathbf{1}_J \mathbf{1}_J')$ with inverse given by (9).

518 3.4.1 Computing $X' \Sigma^{-1} X$

519 [27] The Woodbury computation is organized around the
520 sum-of-squares of the run-in and post-randomization time
521 indices.

- 522 • The derivation proceeds via the Woodbury identity.
- 523 • S_{J_0} is the sum of squared run-in indices in closed form.
- 524 • S_{J_1} is the corresponding sum over post-randomization indices.

525 To be completed by rgt.

526 We proceed through the Woodbury identity. Define $S_{J_0} = \sum_{s=1}^{J_0} s^2 = J_0(J_0 +$
527 $1)(2J_0 + 1)/6$ and $S_{J_1} = \sum_{l=1}^{J_1} l^2 = J_1(J_1 + 1)(2J_1 + 1)/6$.

528 The key intermediate quantities are:

529 $Z' R^{-1} Z$: With $Z = t \cdot \mathbf{d}$ where $\mathbf{d} = (-J_0, \dots, -1, 1, \dots, J_1)'$:

$$Z' R^{-1} Z = \frac{t^2}{a} \left[(S_{J_0} + S_{J_1}) - \frac{(\sum d_j)^2}{J + 1} \right] \quad (32)$$

530 where $\sum d_j = -J_0(J_0 + 1)/2 + J_1(J_1 + 1)/2$.

531 $(G^{-1} + Z'R^{-1}Z)^{-1}$: From (32):

$$H = \frac{ab}{a + t^2b \cdot Z'R^{-1}Z \cdot a/t^2}$$

532 $X'R^{-1}X$: This 3×3 matrix has elements computed from the block structure of
 533 X . Letting $D_{J_0} = \sum_{s=1}^{J_0} s = J_0(J_0 + 1)/2$ and $D_{J_1} = \sum_{l=1}^{J_1} l = J_1(J_1 + 1)/2$, and
 534 using $R^{-1} = \frac{1}{a}(\mathbf{I}_J - \frac{1}{J+1}\mathbf{1}\mathbf{1}')$, the information matrix summed over one treatment
 535 and one placebo participant is:

$$X'R^{-1}X = \frac{t^2}{a} \begin{pmatrix} 2\left(S_{J_0} - \frac{D_{J_0}^2}{J+1}\right) & \frac{2D_{J_0}D_{J_1}}{J+1} & \frac{D_{J_0}D_{J_1}}{J+1} \\ \frac{2D_{J_0}D_{J_1}}{J+1} & 2\left(S_{J_1} - \frac{D_{J_1}^2}{J+1}\right) & S_{J_1} - \frac{D_{J_1}^2}{J+1} \\ \frac{D_{J_0}D_{J_1}}{J+1} & S_{J_1} - \frac{D_{J_1}^2}{J+1} & S_{J_1} - \frac{D_{J_1}^2}{J+1} \end{pmatrix} \quad (33)$$

536 [28] **The structure of the information matrix reflects which**
 537 **parameters both groups inform and the opposing signs of**
 538 **the two phase time-sums.**

- 539 • Entries informed by both groups carry a factor of 2.
- 540 • The γ -related entries lack this factor because the placebo γ col-
 541 umn is zero.
- 542 • The positive (δ, β) cross-term arises from the opposite signs of
 543 the run-in and post-randomization time sums.

544 *To be completed by rgt.*

545 *The factor of 2 appears on entries involving both groups, while γ -related*
 546 *entries lack this factor since the γ column is zero for the placebo group. The*
 547 *positive sign on the (δ, β) cross-term arises because the run-in and post-*
 548 *randomization time sums have opposite signs ($\sum d_{J_0} < 0$, $\sum d_{J_1} > 0$).*

549 **Woodbury correction:** The correction is the sum of per-subject rank-one outer
 550 products:

$$W = \sum_{g \in \{\text{trt}, \text{plc}\}} X_i^{(g)'} R^{-1} Z \cdot H \cdot Z' R^{-1} X_i^{(g)}$$

551 which is generally rank 2 (not rank 1), so $X'\Sigma^{-1}X = X'R^{-1}X - W$.

552 3.4.2 Numerical implementation

553 [29] **For general run-in and post-randomization counts the**
 554 **variance has no compact closed form and must be evaluated**
 555 **as a complex ratio.**

- 556 • No simple closed-form expression exists for arbitrary J_0 and J_1 .
- 557 • The variance depends on the sum-of-squares and sum quantities
 558 together with the dimension J .
- 559 • The Woodbury correction interacts with the information matrix
 560 to produce algebraically complex expressions.

561 To be completed by rgt.

562 *The variance $\text{Var}(\hat{\gamma})$ for general J_0 and J_1 does not admit a simple closed-*
 563 *form expression. The formula depends on σ^2 and σ_b^2 in a ratio involving S_{J_0} ,*
 564 *S_{J_1} , D_{J_0} , D_{J_1} , and J , and the interaction between the Woodbury correction*
 565 *and the information matrix produces algebraically complex expressions.*

566 [30] **A matrix-based R implementation evaluates the exact**
 567 **variance for any configuration, with the closed-form cases as**
 568 **checkpoints.**

- 569 • The computation uses the matrix Woodbury identity in R code.
- 570 • It returns the exact variance for any (J_0, J_1, J_2) configuration.
- 571 • The $J_0 = 0$ and $J_0 = 2$ closed forms serve as verification check-
- 572 points.

573 To be completed by rgt.

574 *We therefore implement the computation via the matrix-based Woodbury iden-*
 575 *tity in R code, which evaluates the exact variance for any configuration*
 576 *(J_0, J_1, J_2) . The closed-form expressions for $J_0 = 0$ (19) and $J_0 = 2$ (30)*
 577 *serve as verification checkpoints.*

578 3.4.3 Remark: recovery of the Edland and Diggle formulae

579 [31] **Setting the run-in count to zero collapses the informa-**
 580 **tion matrix to a two-parameter block and yields the slope-**
 581 **comparison variance.**

- 582 • With $J_0 = 0$ the δ column vanishes, leaving a 2×2 block on
- 583 (β, γ) .
- 584 • Sherman-Morrison cancellations collapse the Woodbury correc-
- 585 tion against the remaining structure.
- 586 • The result is the per-arm variance of the treatment-effect estima-
- 587 tor for n participants.

588 To be completed by rgt.

589 *At $J_0 = 0$, the δ column of X is empty and the information matrix reduces*
 590 *to a 2×2 block on (β, γ) . Carrying the Woodbury algebra through, the*
 591 *Sherman-Morrison cancellations collapse the Woodbury correction against*
 592 *the $(2, 1; 1, 1)$ structure of $X'R^{-1}X$, and the variance of the treatment effect*
 593 *estimator for n participants per arm becomes*

$$\text{Var}(\hat{\gamma})|_{J_0=0, J_2=0} = \frac{2}{n} \left[\sigma_b^2 + \frac{\sigma^2}{S_{xx}} \right], \quad S_{xx} = \sum_{j=0}^{J_1} (t_j - \bar{t})^2, \quad (34)$$

594 [32] **The boundary case recovers the Edland, Diggle, and Frost**
 595 **formulae, confirming that the general result embeds the long-**
 596 **power routines as strict special cases.**

- The expression is the Ard-Edland slope-comparison formula implemented in the `longpower edland` routine.
- Setting the slope variance to zero gives compound symmetry and recovers the Diggle formula.
- The Frost formula is the two-visit specialization, so the general result quantifies gains accruing only from run-in or common-close observations.

To be completed by rgt.

where the baseline visit at $t_0 = 0$ enters through the change-score transformation. Equation (34) is the slope-comparison formula of⁸ implemented in `longpower::edland.linear.power()`. Setting additionally $\sigma_b^2 = 0$ reduces Σ to compound symmetry and recovers the formula of⁹ used by `longpower::diggle.linear.power()`. The Frost formula (19) is in turn the $J_1 = 2$ specialization of (34), since $S_{xx} = 2t^2$ at that setting. The general- (J_0, J_1, J_2) result derived in the next subsection therefore embeds both `longpower` closed-form routines as strict boundary cases, with the efficiency gains quantified in Section 5 accruing entirely from the information contributed by $J_0 > 0$ or $J_2 > 0$.

3.5 General formula with common close ($J_2 \geq 0$)

[33] **The final extension adds common-close observations after the treatment period to the change-score vector.**

- J_2 common-close observations follow the treatment period.
- All participants are off-treatment during this phase.
- The total dimension becomes $J = J_0 + J_1 + J_2$.

To be completed by rgt.

We now extend to include J_2 common close observations after the treatment period. The total dimension is $J = J_0 + J_1 + J_2$.

During the common close phase (time points $J_1 + 1, \dots, J_1 + J_2$), all participants are off-treatment. The change scores are:

$$c_{i,J_1+s} = [\beta(J_1 + s) + \gamma g_i \cdot J_1 + b_i(J_1 + s)]t + (w_{i,J_1+s} - w_{i0}) \quad s = 1, \dots, J_2 \quad (35)$$

[34] **In the common close the treatment effect persists only as an accumulated offset, fixing the design-matrix entry at the treatment-period value.**

- γ enters only through the effect accumulated during the treatment period.
- It does not contribute to the slope during the common close.
- The γ column therefore equals $J_1 t$ for all common-close observations in the treatment group.

634 To be completed by rgt.

635 *The treatment effect γ enters the common close change scores only through*
 636 *the accumulated effect during the treatment period ($\gamma g_i \cdot J_1 \cdot t$), not through*
 637 *the slope during the common close. This means the γ column in the design*
 638 *matrix has the value $J_1 t$ for all common close observations in the treatment*
 639 *group.*

640 The fixed effects design matrix becomes:

$$X_i^{(\text{trt})} = t \begin{pmatrix} -J_0 & 0 & 0 \\ \vdots & \vdots & \vdots \\ -1 & 0 & 0 \\ 0 & 1 & 1 \\ \vdots & \vdots & \vdots \\ 0 & J_1 & J_1 \\ 0 & J_1 + 1 & J_1 \\ \vdots & \vdots & \vdots \\ 0 & J_1 + J_2 & J_1 \end{pmatrix}$$

641 [35] **The slope and treatment columns of the design matrix**
 642 **encode the post-randomization times and the capped treat-**
 643 **ment exposure respectively.**

- 644 • The β column holds the post-randomization times of the placebo
 645 slope across all later visits.
- 646 • The γ column rises through the treatment period then stays at
 647 J_1 during the common close.
- 648 • The γ column is identically zero for placebo participants.

649 To be completed by rgt.

650 *The β column contains the post-randomization time for the placebo slope:*
 651 *$0, \dots, 0, 1, \dots, J_1, J_1 + 1, \dots, J_1 + J_2$. The γ column is $0, \dots, 0, 1, \dots, J_1, J_1, \dots, J_1$*
 652 *for treatment participants and all zeros for placebo.*

653 The random effects design remains $Z_i = t \cdot \mathbf{d}$ where $\mathbf{d} = (-J_0, \dots, -1, 1, \dots, J_1, J_1 +$
 654 $1, \dots, J_1 + J_2)'$.

655 The residual covariance is still $R = a(\mathbf{I}_J + \mathbf{1}_J \mathbf{1}_J')$ with $J = J_0 + J_1 + J_2$.

656 [36] **The common-close extension reuses the same Woodbury**
 657 **computation with only three quantities altered.**

- 658 • The dimension J increases to include the common-close visits.
- 659 • The X matrix changes only in its γ column.
- 660 • The elements of the random-effects vector Z are extended accord-
- 661 **ingly.**

662 To be completed by rgt.

663 *The Woodbury computation proceeds identically, with the only changes being*
 664 *the dimension J , the structure of the X matrix (specifically the γ column),*

665 and the elements of Z .

666 We implement the general computation in R below.

667 4 Power and Sample Size Formulas

668 For a two-sided test at significance level α with power $1 - \beta_{\text{pow}}$, the required
669 number of participants per group is

$$m = \frac{(z_{\alpha/2} + z_{1-\beta_{\text{pow}}})^2 \cdot \text{Var}_1(\hat{\gamma})}{\Delta^2} \quad (36)$$

670 where $\text{Var}_1(\hat{\gamma})$ denotes the per-participant-pair variance from the formulas derived
671 above, and Δ is the minimum clinically important treatment effect on the rate of
672 change.

673 Equivalently, the power for a given sample size is

$$1 - \beta_{\text{pow}} = \Phi\left(\frac{\Delta\sqrt{m}}{\sqrt{\text{Var}_1(\hat{\gamma})}} - z_{\alpha/2}\right) \quad (37)$$

674 The relative efficiency of a design with (J_0, J_1, J_2) observations compared to a
675 standard design with $(0, J_1, 0)$ is

$$\text{RE}(J_0, J_1, J_2) = \frac{\text{Var}_1(\hat{\gamma})|_{0, J_1, 0}}{\text{Var}_1(\hat{\gamma})|_{J_0, J_1, J_2}} \quad (38)$$

676 Values greater than 1 indicate that the run-in/common close design requires fewer
677 participants.

678 **[37] A single run-in observation can be counterproductive be-**
679 **cause the extra slope parameter it introduces may offset the**
680 **information it contributes.**

- 681 • Adding one run-in observation introduces the parameter δ , en-
682 larging the fixed-effect dimension.
- 683 • This added degree of freedom can make the one-run-in variance
684 exceed the no-run-in variance for some parameters.
- 685 • Designers should therefore plan either zero or at least two run-in
686 observations, never exactly one.

687 To be completed by rgt.

688 *A practical caveat: adding a single run-in observation ($J_0 = 1$) introduces the*
689 *run-in slope parameter δ into the model, increasing the dimension of β from*
690 *2 to 3. This additional degree of freedom can offset the information gained*
691 *from the run-in observation, so that $\text{Var}(\hat{\gamma})|_{J_0=1}$ may exceed $\text{Var}(\hat{\gamma})|_{J_0=0}$ for*
692 *some parameter configurations. This effect, noted in a different context by¹⁵,*

693 *implies that trial designers should plan either zero or at least two run-in*
694 *observations, not one.*

695 5 Numerical Examples

696 5.1 Verification of closed-form results

697 [38] **The closed-form expressions are validated against the ma-**
698 **trix computation in two representative design configurations.**

- 699 • The verification confirms agreement between the analytic and
700 matrix-based variances.
- 701 • The Frost standard design uses the two-parameter model with
702 two post-baseline visits.
- 703 • The two-run-in case exercises the three-parameter model.

704 To be completed by rgt.

705 *We verify that the closed-form expressions agree with the matrix computation.*
706 *The Frost et al. (2008) result corresponds to the standard design with $J_0 = 0$,*
707 *$J_1 = 2$ (two post-baseline visits at times t and $2t$), which under the two-*
708 *parameter model yields $\text{Var}(\hat{\gamma}) = (\sigma^2 + 2\sigma_b^2 t^2)/t^2$. The $J_0 = 2$, $J_1 = 1$ case*
709 *uses the three-parameter model.*

```

a <- 1.0
b <- 0.5
tt <- 1.0

v_frost_matrix <- var_gamma_matrix(0, 2, 0, a, b, tt)
v_frost_closed <- var_gamma_frost(a, b, tt)

v_r2_matrix <- var_gamma_matrix(2, 1, 0, a, b, tt)
v_r2_closed <- var_gamma_r2(a, b, tt)

cat('Frost (r=0, k=2) case:\n')

710 ## Frost (r=0, k=2) case:

cat('  Matrix:', round(v_frost_matrix, 6), '\n')

711 ##   Matrix: 2

cat('  Closed:', round(v_frost_closed, 6), '\n')

712 ##   Closed: 2

cat('  Match:',
    all.equal(v_frost_matrix, v_frost_closed), '\n\n')

713 ##   Match: TRUE

```

Table 1: Required sample size per group ($J_1 = 2$, $J_2 = 0$, $\Delta = 0.5$, 90% power, two-sided $\alpha = 0.05$) as a function of the number of run-in observations J_0 , for three levels of between-subject slope variability.

J_0	Low	Medium	High
0	51	127	463
1	44	119	214
2	44	91	113
3	43	70	76
4	42	56	59
5	40	48	49
6	38	42	43

```

cat('r=2, k=1 case:\n')

714 ## r=2, k=1 case:
cat(' Matrix:', round(v_r2_matrix, 6), '\n')

715 ## Matrix: 4.666667
cat(' Closed:', round(v_r2_closed, 6), '\n')

716 ## Closed: 4.666667
cat(' Match:',
    all.equal(v_r2_matrix, v_r2_closed), '\n\n')

717 ## Match: TRUE
cat('Variance comparison across designs:\n')

718 ## Variance comparison across designs:
for (rr in 0:4) {
  v <- var_gamma_matrix(rr, 2, 0, a, b, tt)
  cat(sprintf(' r=%d, k=2: Var = %.4f\n', rr, v))
}

719 ## r=0, k=2: Var = 2.0000
720 ## r=1, k=2: Var = 2.0000
721 ## r=2, k=2: Var = 1.7910
722 ## r=3, k=2: Var = 1.5000
723 ## r=4, k=2: Var = 1.2651

```


Table 2: Required sample size per group ($J_0 = 2$, $J_1 = 2$, $\Delta = 0.5$, 90% power) as a function of common close observations J_2 .

J_2	Low	Medium	High
0	44	91	113
1	34	69	80
2	29	48	52
3	25	35	37
4	22	27	28

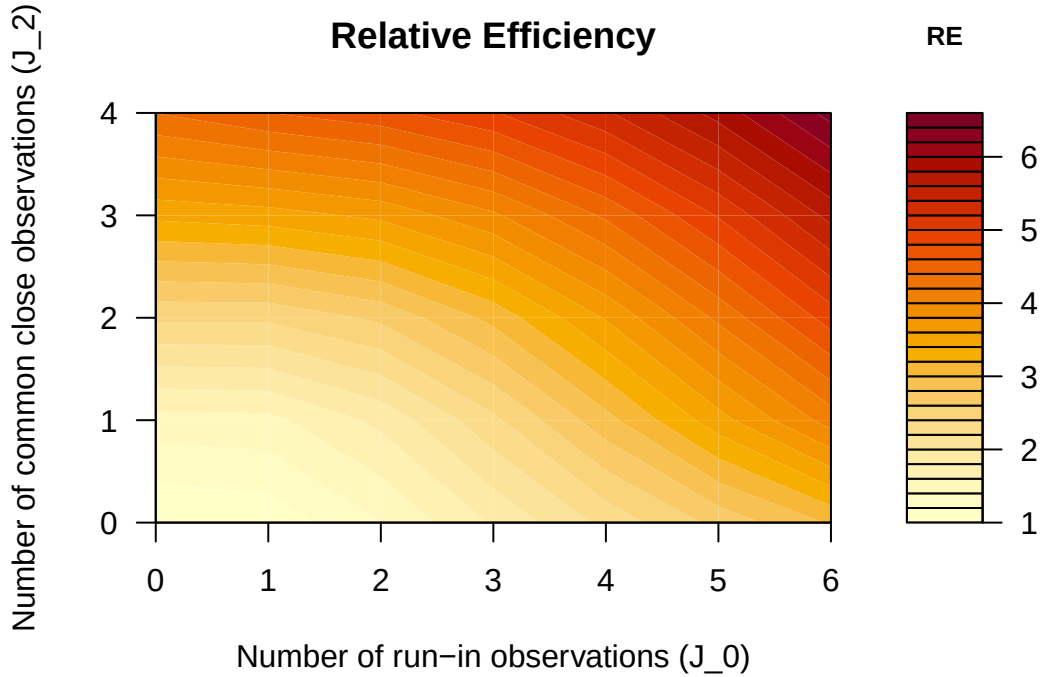


Figure 1: Relative efficiency of run-in and common close designs compared to the standard design ($J_0 = 0$, $J_2 = 0$) with the same number of post-randomization observations ($J_1 = 2$). Parameters: $\sigma^2 = 1$, $\sigma_b^2 = 1$, $t = 1$.

724 5.2 Sample size as a function of run-in observations

725 5.3 Sample size as a function of common close observations

726 5.4 Relative efficiency surface

727 5.5 Application: Alzheimer’s disease imaging trial

728 [39] The Alzheimer’s application uses MIRIAD-derived pa-
729 rameters with the correctly combined within-visit residual,
730 which is essential to reproduce Frost’s reported sample sizes.

- 731 • The slope variance and combined within-visit residual are taken
732 from Frost’s MIRIAD estimates.
- 733 • The example assumes a one-year observation interval and a treat-
734 ment effect of one quarter.
- 735 • Using only the small measurement-error term understates the
736 residual by two orders of magnitude and yields degenerate sample
737 sizes.

738 To be completed by rgt.

739 *Using variance parameters estimated from the MIRIAD dataset²: $\sigma_b^2 =$
740 0.9745 (between-subject slope variance) and a combined within-visit resid-
741 ual $\sigma^2 = \sigma_u^2 + \sigma_\varepsilon^2 = 0.3338 + 0.0025 = 0.3363$, formed per Eq. (2) from
742 the within-subject between-visit component ($\sigma_u^2 = 0.3338$) and the additional
743 between-scan measurement error ($\sigma_\varepsilon^2 = 0.0025$) of Frost’s Table II. We take
744 observation interval $t = 1$ year and treatment effect $\Delta = 0.25$ (25% re-
745 duction in atrophy rate relative to placebo rate of approximately 1.0% brain
746 volume loss per year). Using the combined residual, the conventional design
747 reproduces Frost’s reported sample sizes (Table III); using only the 0.0025
748 measurement-error term, as in earlier drafts, understates the residual by two
749 orders of magnitude and yields degenerate sample sizes.*

750 ## AD imaging trial sample sizes (per group):

751 ## Standard (r=0, k=2, f=0): 385

752 ## Run-in (r=2, k=2, f=0): 145 (62% reduction)

753 ## Full (r=2, k=2, f=2): 69 (82% reduction)

754 6 Discussion

755 [40] The work formalizes and extends Frost’s framework with
756 explicit variance formulas for run-in and common-close de-
757 signs.

- 758 • The results provide explicit treatment-effect variance formulas.
- 759 • They cover designs incorporating both run-in and common-close
760 observations.
- 761 • Three substantive points warrant further discussion.

Table 3: Sample size per group for AD imaging trial ($J_1 = 2$ post-randomization visits, $\Delta = 0.25$, 90% power). Top 15 configurations by sample size.

J_0	J_1	J_2	m per group	Total visits
4	2	2	49	9
3	2	2	58	8
4	2	1	60	8
2	2	2	69	7
3	2	1	77	7
0	2	2	77	5
1	2	2	77	6
4	2	0	78	7
3	2	0	100	6
2	2	1	104	6
1	2	1	138	5
2	2	0	145	5
0	2	1	150	4
1	2	0	247	4

To be completed by rgt.

The results presented here formalize and extend the framework of² by providing explicit variance formulas for the treatment effect estimator under designs that incorporate run-in and common close observations. Three points merit discussion.

[41] **The benefit of run-in observations is governed by the slope-to-noise ratio and is largest for outcomes such as neuroimaging.**

- The efficiency gain depends on the ratio of slope variance to residual variance.
- When that ratio is large, even one or two run-in observations cut sample sizes substantially.
- For cognitive outcomes with a less favorable ratio, gains are modest and must be weighed against longer study duration.

To be completed by rgt.

When run-in observations are most beneficial. The efficiency gain from run-in observations depends on the ratio $\sigma_b^2 t^2 / \sigma^2$. When between-subject slope variability is large relative to residual variance (as in neuroimaging outcomes), even one or two run-in observations yield substantial sample size reductions. For cognitive outcomes, where this ratio is typically less favorable, the gains are more modest and the cost of extended study duration must be weighed against the sample size reduction.

[42] **Common-close observations sharpen the treatment-effect estimator at low cost, with diminishing returns beyond the**

786

first few visits.

787

788

789

790

791

792

793

To be completed by rgt.

794

795

796

797

798

799

800

801

Common close as a low-cost efficiency gain. The common close phase provides information about individual trajectories without requiring participants to remain on treatment. Since treatment has ceased, both groups contribute to estimation of the underlying slope variability, effectively tightening the precision of the treatment effect estimator. The marginal benefit of each additional common close observation diminishes (as with run-in observations), but the first one or two common close visits typically provide meaningful sample size reductions, particularly when $\sigma_b^2 t^2 / \sigma^2$ is large.

802

803

804

[43] **The derived formulas are consistent with Wang’s two-period model while making the three phases explicit and separately optimizable.**

805

806

807

808

809

810

- Wang’s common-slope scenario corresponds to setting the run-in and placebo slopes equal.
- Their different-slope scenario corresponds to the general case treated here.
- The present formulation separates run-in, treatment, and common-close phases, enabling per-phase optimization.

811

To be completed by rgt.

812

813

814

815

816

817

818

Relationship to the two-period model. The formulas derived here are consistent with the two-period linear mixed effects model of³. Their Scenario 1 (common slope across periods) corresponds to setting $\delta = \beta$ in our model, while their Scenario 2 (different slopes) corresponds to the general case. The advantage of our formulation is the explicit separation of run-in, treatment, and common close phases, enabling direct optimization of the number of observations in each phase.

819

820

821

[44] **The derivations rest on simplifying assumptions whose relaxation is feasible but typically forfeits closed-form simplicity.**

822

823

824

825

826

- They assume equally spaced observations, linear trajectories, and a random intercept-plus-slope structure.
- Extensions to unequal spacing, nonlinear progression, or richer random effects remain within the Woodbury framework but lose simple closed forms.

- No dropout is assumed, though the Dawson-Lagakos pattern-mixture approach can adjust for anticipated attrition.

To be completed by rgt.

Limitations. The derivations assume equally spaced observations, linear trajectories, and a random intercept plus random slope covariance structure. Extensions to unequal spacing, nonlinear progression, and more complex random effects structures (e.g., random intercept-slope correlation) are possible within the same Woodbury identity framework but yield more complex expressions that may not admit simple closed forms. The formulas also assume no dropout; in practice, the Dawson-Lagakos pattern-mixture approach² can be applied to adjust for anticipated attrition.

7 Conflict of interest

The author declares no conflicts of interest.

8 Funding

This work received no specific funding.

9 Data availability statement

[45] All artefacts supporting the paper are openly available in the project repository.

- These include the R package, the four companion manuscripts, and the MATLAB derivation scripts.
- The shared bibliography is likewise available.
- Specific paths are enumerated in the Reproducibility section.

To be completed by rgt.

The *runinpower* R package, the four companion manuscripts of this compendium, the MATLAB symbolic-derivation scripts, and the shared bibliography are openly available in the project repository. Specific paths are listed in the Reproducibility section below.

10 Reproducibility

[46] The manuscript build relies on the in-package source together with a small set of testing and rendering packages.

- Rendering used R version 4.4.x.
- The principal dependencies are the in-package source plus `tinytest`, `rmarkdown`, and `bookdown`.
- Supporting MATLAB symbolic derivations reside in the analysis scripts directory.

862 To be completed by rgt.

863 *This manuscript was rendered with R 4.4.x. Principal packages used*
864 *are the in-package runinpower source (nine source files: allocation,*
865 *ancova, averaged, covariance, covariance-ar1, design, power, replicated,*
866 *variance), tinytest (testing harness), and rmarkdown plus bookdown*
867 *for the manuscript build. MATLAB symbolic derivations supporting the*
868 *Woodbury-based variance formulas are at analysis/scripts/.*

869 [47] **The variance results are deterministic, and any Monte**
870 **Carlo benchmarking uses a fixed seed for reproducibility.**

- 871 • Given fixed input parameters the formulas exercise no random-
872 number generation.
- 873 • Monte Carlo benchmarking against the analytic formulas is the
874 only stochastic component.
- 875 • That benchmarking driver fixes the seed at its entry point.

876 To be completed by rgt.

877 *Random-number generator. The variance formulas are deterministic*
878 *given fixed input parameters; no RNG is exercised. Where Monte Carlo*
879 *benchmarking against the analytic formulas is reported, the driver sets*
880 *set.seed(20260418) at its entry point.*

881 [48] **The repository organizes the package source, derivation**
882 **scripts, companion manuscripts, and bibliography into well-**
883 **defined locations.**

- 884 • The R package source and the MATLAB scripts each have a
885 dedicated directory.
- 886 • The three companion manuscripts share a report subtree.
- 887 • This manuscript source and the shared bibliography sit in the
888 run-in power report directory.

889 To be completed by rgt.

890 *Data and code availability. The R package source is at R/. MAT-*
891 *LAB scripts are at analysis/scripts/. The companion manuscripts are at*
892 *analysis/report/{02,03,04}-*/. The manuscript source and shared bibli-*
893 *ography (./references.bib) are at analysis/report/01-runin-power/.*

894 11 References

- 895 1. Packer M. Why has a run-in period been a design element in most
landmark clinical trials? Analysis of the critical role of run-in periods
in drug development. *Journal of Cardiac Failure*. 2017;23(10):697-699.
doi:10.1016/j.cardfail.2017.07.009

- 896 2. Frost C, Kenward MG, Fox NC. Optimizing the design of clinical trials where the outcome is a rate. Can estimating a baseline rate in a run-in period increase efficiency? *Statistics in Medicine*. 2008;27(18):3717-3731. doi:10.1002/sim.3280
- 897 3. Wang G, Aschenbrenner AJ, Li Y, et al. Two-period linear mixed effects models to analyze clinical trials with run-in data when the primary outcome is continuous: Applications to Alzheimer's disease. *Alzheimer's & Dementia: Translational Research & Clinical Interventions*. 2019;5:450-457. doi:10.1016/j.trci.2019.08.002
- 898 4. Lu K, Mehrotra DV, Liu GF. Sample size determination for constrained longitudinal data analysis. *Statistics in Medicine*. 2009;28(4):679-699. doi:10.1002/sim.3507
- 899 5. Lu K. On efficiency of constrained longitudinal data analysis versus longitudinal analysis of covariance. *Biometrics*. 2010;66(3):891-896. doi:10.1111/j.1541-0420.2009.01322.x
- 900 6. Hu N, Mackey H, Thomas R. Power and sample size for random coefficient regression models in randomized experiments with monotone missing data. *Biometrical Journal*. 2021;63(4):806-824. doi:10.1002/bimj.202000184
- 901 7. Iddi S, Donohue MC. Power and sample size for longitudinal models in R – the longpower package and Shiny app. *The R Journal*. 2022;14(1):264-282. <https://journal.r-project.org/articles/RJ-2022-022/>
- 902 8. Ard MC, Edland SD. Power calculations for clinical trials in Alzheimer's disease. *Journal of Alzheimer's Disease*. 2011;21:369-377. doi:10.3233/JAD-2011-0062
- 903 9. Diggle PJ, Heagerty P, Liang KY, Zeger SL. *Analysis of Longitudinal Data*. 2nd ed. Oxford University Press; 2002.
- 904 10. Liu G, Liang KY. Sample size calculations for studies with correlated observations. *Biometrics*. 1997;53:937-947. doi:10.2307/2533554
- 905 11. Sperling RA, Rentz DM, Johnson KA, et al. The A4 study: Stopping AD before symptoms begin? *Science Translational Medicine*. 2014;6(228):228fs13. doi:10.1126/scitranslmed.3007941
- 906 12. Laird NM, Ware JH. Random-effects models for longitudinal data. *Biometrics*. 1982;38(4):963-974. doi:10.2307/2529876

- 907 13. Wang G, Liu L, Li Y, et al. Proportional constrained longitudinal data analysis models for clinical trials in sporadic Alzheimer's disease. *Alzheimer's & Dementia: Translational Research & Clinical Interventions*. 2022;8(1):e12286. doi:10.1002/trc2.12286
- 908 14. Zhao Y, Edland SD. Power formulas for mixed effects models with random slope and intercept comparing rate of change across groups. *The International Journal of Biostatistics*. 2022;18(1):173-182. doi:10.1515/ijb-2020-0107
- 909 15. Frison L, Pocock SJ. Linearly divergent treatment effects in clinical trials with repeated measures: Efficient analysis using summary statistics. *Statistics in Medicine*. 1997;16(24):2855-2872. doi:10.1002/(SICI)1097-0258(19971230)16:24<2855::AID-SIM749>3.0.CO;2-2